

Selenium: A Popular Software Testing Tool You Can Rely On

For the software enthusiast, this article begins with an overview on software testing. The various types of testing and the many software testing tools are enumerated, and then the author goes into detailed discussion on the free and open source automation testing tool, Selenium.



Software testing is a process of evaluating a system or its components to find out whether the specified requirements are met. It can also be referred to as a system to identify any gaps, errors or missing requirements. It comprises verification and validation of the software product. This process starts from the beginning of the software development lifecycle (SDL) till it is delivered to the customer.

The purpose of software testing is to deliver a higher quality of product. If the product produces an error at a later stage, then it becomes costly to remove it. Software testing has the potential to save money and time by identifying problems early and delivering a defect-free product.

It has three main objectives:

- **Verification:** This process confirms that the software meets its technical specifications.
 - **Validation:** This process confirms that the software meets the business requirements.
 - **Defect:** This detects any differences between the expected and actual results.
- Software testing can be done in various ways and is broadly divided into two types:
- **Manual testing:** This type of testing involves software developers running tests manually.
- There are several stages at which manual testing can be done. These are:
- Unit testing
 - Integration testing
 - Software testing
 - Functional testing
 - Non-functional testing
 - System testing